

## **PHASE 3 – PROJECT DESIGN**

### **MODULE 3.3 – SOLUTION ARCHITECTURE**

#### **3.3.1 Introduction**

Solution Architecture describes the major components of the proposed system and explains how they interact with each other.

For the project “**Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer,**” the architecture is based on ServiceNow components that work together to manage the laptop procurement workflow.

#### **3.3.2 Architecture Overview**

The proposed architecture connects the employee, ServiceNow, Flow Designer, approval process, IT/procurement team, and notification system.

The overall architecture can be represented as:

**Employee**



**ServiceNow Service Catalog / Request Form**



**Laptop Request Record**



**Flow Designer**



**Approval Process**



**IT / Procurement Task**



**Laptop Procurement and Fulfilment**



**Request Status Update**



**Notification to Employee**

#### **3.3.3 Major Architecture Components**

### **1. Employee Layer**

The employee is the primary user of the system. The employee submits a request for a standard laptop and provides the required information.

### **2. ServiceNow Request Layer**

ServiceNow provides the interface through which the employee submits the request. The submitted information is stored as a request record.

This layer acts as the entry point for the procurement workflow.

### **3. Flow Designer Layer**

Flow Designer is the main automation component of the proposed system.

After the request is submitted, Flow Designer triggers the predefined workflow and manages the different stages of the process.

It can perform actions such as:

- Starting the workflow.
- Checking request conditions.
- Handling approvals.
- Creating tasks.
- Assigning tasks.
- Updating records.
- Triggering notifications.

### **3. Approval Layer**

The approval layer manages the authorization required before the laptop order can proceed.

The request is reviewed by the appropriate approver. Based on the configured workflow, the process continues after the required approval.

### **5. IT and Procurement Layer**

After approval, the responsible team receives the required task. The team processes the laptop request and performs the necessary procurement activities.

### **6. Data and Record Layer**

ServiceNow records contain the information required to manage the request and its progress.

The records help maintain:

- Request details.
- Approval information.

- Task information.
- Status information.
- Processing history.

## **7. Notification Layer**

The notification component provides updates to the employee and relevant users.

Notifications may be triggered at important stages such as:

- Request submission.
- Approval.
- Processing.
- Completion.

### **3.3.3 Data and Process Flow**

The flow of information through the architecture is:

#### **Employee**

→ Submits standard laptop request

#### **ServiceNow**

→ Stores request information

#### **Flow Designer**

→ Starts the automated workflow

#### **Approval**

→ Request is reviewed

#### **IT/Procurement**

→ Receives and processes the task

#### **ServiceNow**

→ Updates request status

#### **Notification**

→ Employee receives the relevant update

### **3.3.5 Architecture Layers**

The solution can be divided into four main layers:

| <b>Layer</b>      | <b>Main Component</b> | <b>Purpose</b>                 |
|-------------------|-----------------------|--------------------------------|
| User Layer        | Employee              | Submit laptop request          |
| Application Layer | ServiceNow            | Manage request and records     |
| Automation Layer  | Flow Designer         | Automate workflow activities   |
| Processing Layer  | IT/Procurement Team   | Process and fulfil the request |

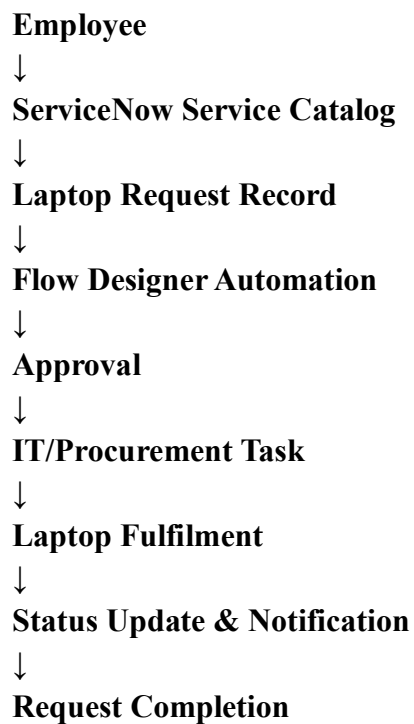
### **3.3.6 Architecture Benefits**

The proposed architecture provides:

- Clear separation of system responsibilities.
- Centralized request management.
- Automated workflow execution.
- Structured approval handling.
- Better task management.
- Improved status visibility.
- Easier process monitoring.
- Reduced manual dependency.

### **3.3.7 Overall Architecture**

The overall solution architecture can be summarized as:



### 3.3.8 Conclusion

The solution architecture provides a structured view of how the different components of the project interact.

ServiceNow acts as the central platform, while Flow Designer provides the automation required to move the laptop request through approval, task processing, fulfilment, and completion.

This architecture supports the main objective of the project: **streamlining standard laptop procurement through workflow automation.**